

Practical 12 - Non-linear effects and splines

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `SAheart.csv`

In this practical we will explore modelling non-linear covariate effects using the same South Africa data as the previous two practicals.

Question 1

Load the data, and fit a simple logistic model for `chd` with `sbp` as the single covariate, saving the model for later comparison:

```
SAheart <- read.csv("SAheart.csv", row.names = 1)
chd_linear <- glm(chd~sbp,data=SAheart,family="binomial")
summary(chd_linear)
```

What do the model results show?

SOLUTIONS

```
SAheart <- read.csv("SAheart.csv", row.names = 1)
chd_linear <- glm(chd~sbp,
                  data=SAheart,
                  family="binomial")
summary(chd_linear)
```

Call:

```
glm(formula = chd ~ sbp, family = "binomial", data = SAheart)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.352716	0.687698	-4.875	1.09e-06 ***
sbp	0.019509	0.004863	4.012	6.02e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 596.11 on 461 degrees of freedom
Residual deviance: 579.32 on 460 degrees of freedom
AIC: 583.32

Number of Fisher Scoring iterations: 4

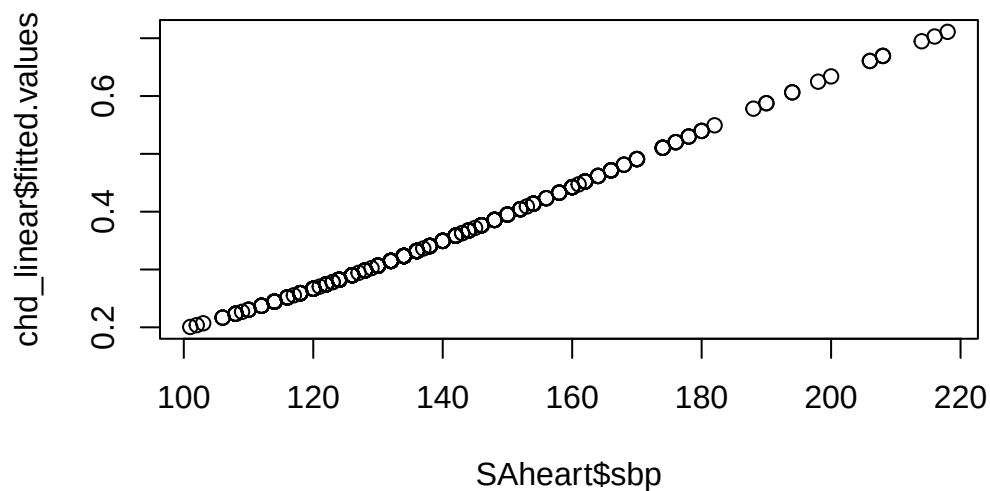
The p-value is highly statistically significant - we have strong evidence of an association, with increasing SBP associated with higher log odds of CHD.

Question 2

Plot the fitted probability values from `chd_linear` against `sbp`. Can you explain what you see?

SOLUTIONS

```
plot(SAheart$sbp, chd_linear$fitted.values)
```

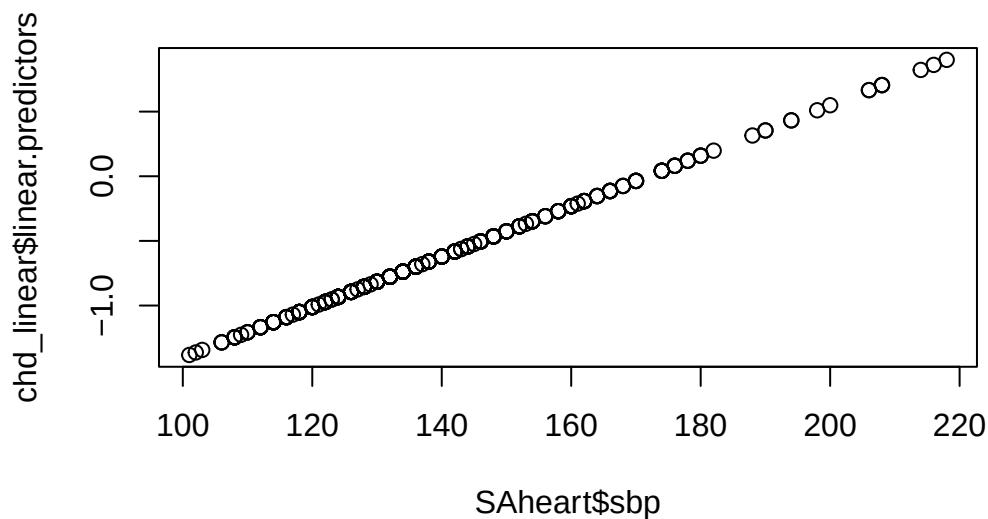


The plot shows the fitted probabilities increasing with increasing SBP, as we should expect given that the coefficient of `sbp` in the model is positive. However, we can see that the function is not exactly linear, even though we included `sbp` linearly in the model.

This is because of the logit link function in logistic regression. This means that $P(CHD = 1|sbp) = \frac{e^{\beta_0 + \beta_1 sbp}}{1 + e^{\beta_0 + \beta_1 sbp}}$, which is not a linear function of `sbp`. Indeed, remember that when the linear predictor gets very large and positive, the probability goes to 1, and when it is very large but negative, the probability goes to 0.

If we instead plot the fitted linear predictor value against `sbp`, we see an exact linear relationship, as we should:

```
plot(SAheart$sbp, chd_linear$linear.predictors)
```



Question 3

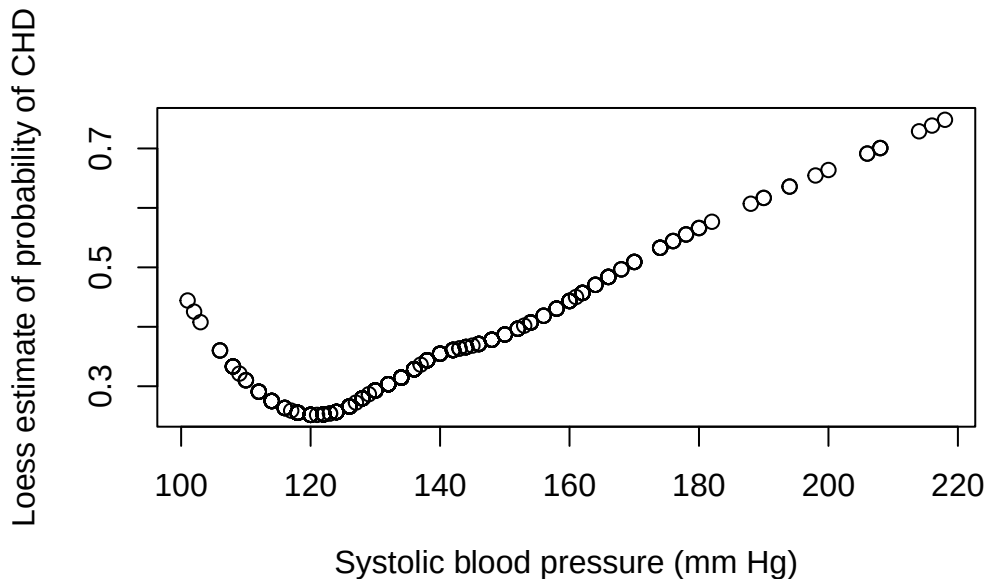
Use the following code to generate a variable containing the loess smoother estimate of the probability of CHD at each value of SBP in the dataset:

```
chd_loess <- loess(chd~sbp,data=SAheart)
```

Next, use the `plot` function to plot `chd_loess$fitted` against `sbp`. What does the loess smoother suggest about the effect of SBP on CHD?

SOLUTIONS

```
chd_loess <- loess(chd~sbp,
                  data=SAheart)
plot(SAheart$sbp, chd_loess$fitted,
     xlab = "Systolic blood pressure (mm Hg)",
     ylab = "Loess estimate of probability of CHD")
```



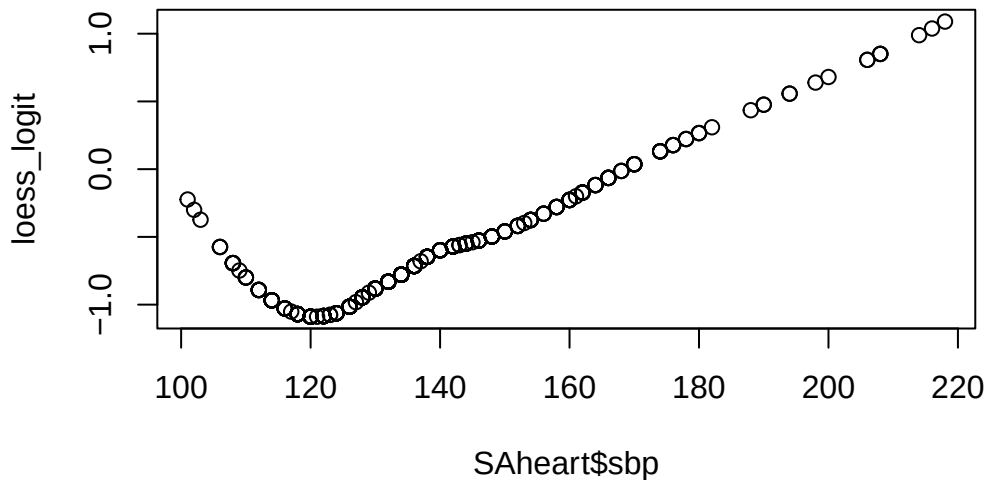
The plot suggests the effect of SBP on the probability of CHD is not monotonic (increasing or decreasing), which is not compatible with our earlier model's assumption of a linear effect on the log odds.

Question 4

Generate a new variable `loess_logit` which takes the values in `chd_loess$fitted` and converts the probabilities to log odds. Then plot `loess_logit` against `sbp`. How has the plot changed compared to the previous question?

SOLUTIONS

```
loess_logit <- log(chd_loess$fitted/(1-chd_loess$fitted))
plot(SAheart$sbp, loess_logit)
```



The plot looks fairly similar to the previous plot, even though the previous one plotted the probability against `sbp` whereas we are now plotting the log odds against `sbp`. They look similar because the log odds transformation we have just applied to the probabilities is close to being linear when the probabilities are not too close to either 0 or 1.

Question 5

Fit a logistic regression for the `chd` outcome with `sbp` and the square of `sbp` as covariates. To do this you can generate a new variable in the data frame for the square term, or include `I(sbp^2)` as a covariate in your model formula.

From your fitted model what do you conclude now about the effect of SBP?

Use a plot to compare the fitted log odds values from this model with the ones estimated using `loess`.

SOLUTIONS

```
chd_quad <- glm(chd~sbp+I(sbp^2),data=SAheart,family="binomial")
summary(chd_quad)
```

Call:

```
glm(formula = chd ~ sbp + I(sbp^2), family = "binomial", data = SAheart)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.7082120	3.6384656	-0.195	0.846
sbp	-0.0164727	0.0489991	-0.336	0.737

```
I(sbp^2)      0.0001195  0.0001625   0.736    0.462
```

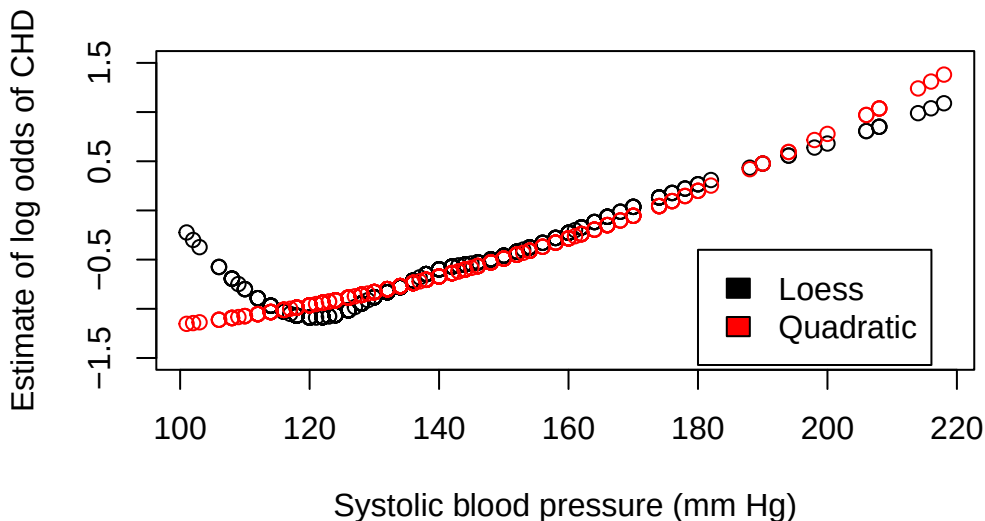
(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 596.11  on 461  degrees of freedom
Residual deviance: 578.77  on 459  degrees of freedom
AIC: 584.77
```

Number of Fisher Scoring iterations: 4

The p-value for the quadratic term is $p = 0.46$ - there is no evidence that the quadratic model fits better than the model that assumed a linear effect (on the log odds scale).

```
plot(SAheart$sbp, loess_logit,
     xlab = "Systolic blood pressure (mm Hg)",
     ylab = "Estimate of log odds of CHD",
     ylim=c(-1.5,1.5))
points(SAheart$sbp, chd_quad$linear.predictors,
       col="red")
legend(180,-0.4, legend=c("Loess", "Quadratic"),
       fill=c("black","red"))
```



We see that the fitted quadratic model has not captured then up turn in log odds on the left hand side that we see in the loess smoother. This is because of the global nature of the quadratic - the coefficient of SBP squared controls the curvature throughout the whole range.

Question 6

Use the following code to fit a logistic model for `chd` with a restricted cubic spline with 4 degrees of freedom for the effect of `sbp`:

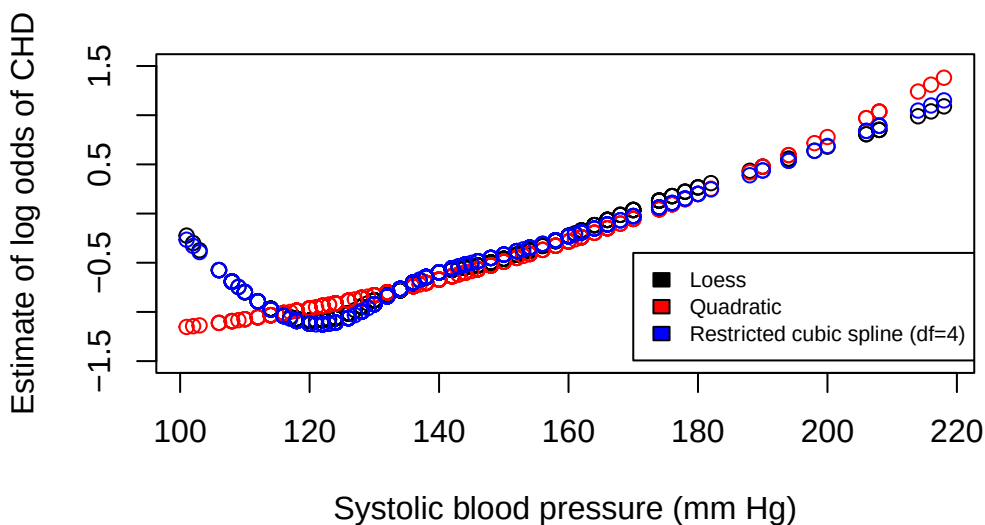
```
library(splines)
chd_rcs <- glm(chd ~ ns(sbp,df=4),data = SAheart,family = binomial)
```

Then create a plot to compare the fitted log odds with the loess and quadratic model estimates.

SOLUTIONS

```
library(splines)
chd_rcs <- glm(chd ~ ns(sbp,df=4),data = SAheart,family = binomial)

plot(SAheart$sbp, loess_logit,
     xlab = "Systolic blood pressure (mm Hg)",
     ylab = "Estimate of log odds of CHD",
     ylim=c(-1.5,1.5))
points(SAheart$sbp, chd_quad$linear.predictors,
       col="red")
points(SAheart$sbp, chd_rcs$linear.predictors,
       col="blue")
legend(170,-0.4, legend=c("Loess", "Quadratic",
                          "Restricted cubic spline (df=4)"),
      fill=c("black","red", "blue"),cex=0.7)
```

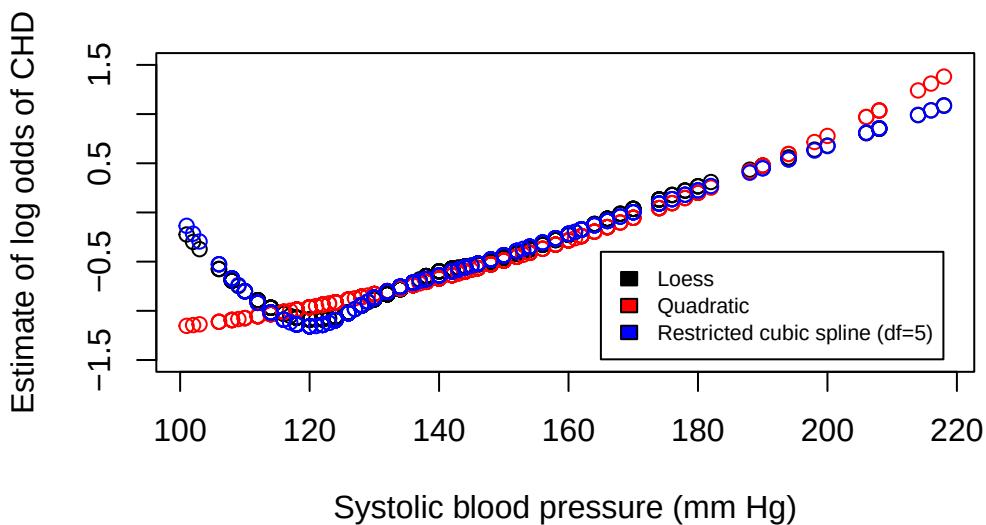
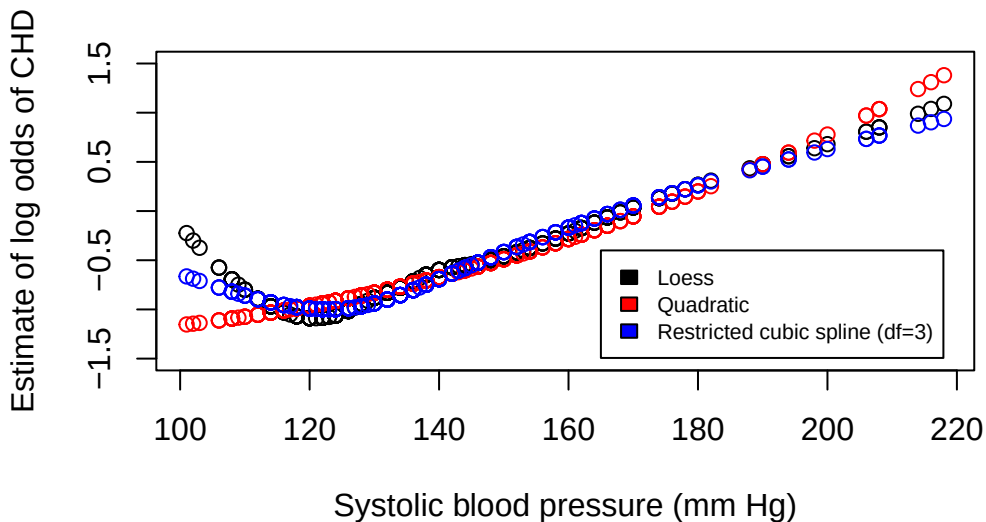


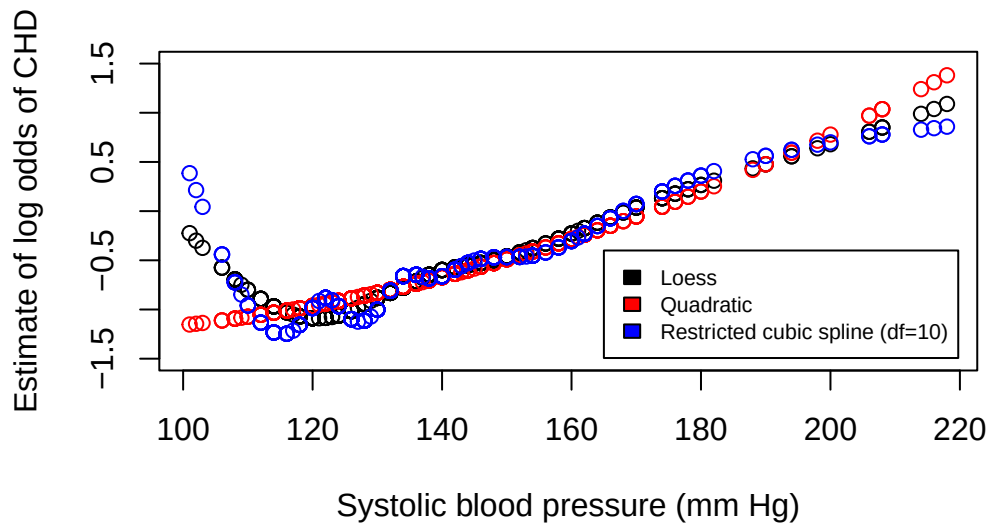
Question 7

Re-run your code from question 6 but with either smaller values of `df` or larger values. Does changing the value of `df` change the fitted log odds in the way you would expect?

SOLUTIONS

In the code below we use the code before and wrap it in a function to make it easier to call with different `df` values:





With `df=3`, the model's fitted log odds values go closer to the loess estimate than the quadratic, but not as closely as our earlier `df=4` model. The model with `df=5` closely follows the loess estimates. But if we increase to `df=10` we overfit and have a fitted function which (almost certainly) wiggles more than would plausibly be true.

Question 8

Use the `anova` function to compare your earlier quadratic effect model with the restricted cubic spline model with `df=4`. What do you conclude?

Is it valid to compare these two models with a likelihood ratio test?

Use the `AIC` function to compare the two models with the `AIC`.

SOLUTIONS

```
anova(chd_quad, chd_rcs)
```

Analysis of Deviance Table

Model 1: `chd ~ sbp + I(sbp^2)`

Model 2: `chd ~ ns(sbp, df = 4)`

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	459	578.77			
2	457	576.85	2	1.9204	0.3828

We are given a p-value of 0.70 - no evidence that the more complex restricted cubic spline model fits better. Remember though that hypothesis tests may fail to reject not because the null is true, but because the power of the test is low. Here in particular we have a moderately sized dataset with a binary outcome, so the power of this test is likely not that high.

The likelihood ratio test is valid if the simpler model is nested within the more complex model. Here this is not the case. One way of seeing why is that the restricted cubic spline model assumes linearity before the first knot and after the last knot, whereas the quadratic model will be curved (not linear) throughout the entire range of `sbp` values.

The AIC can still be used to compare the two models, albeit it does not give a p-value:

```
AIC(chd_quad, chd_rcs)
```

	df	AIC
chd_quad	3	584.7695
chd_rcs	5	586.8491

Models with smaller AIC values are preferred, so here we see the quadratic model is preferred (according to the AIC at least) to the restricted cubic spline model.

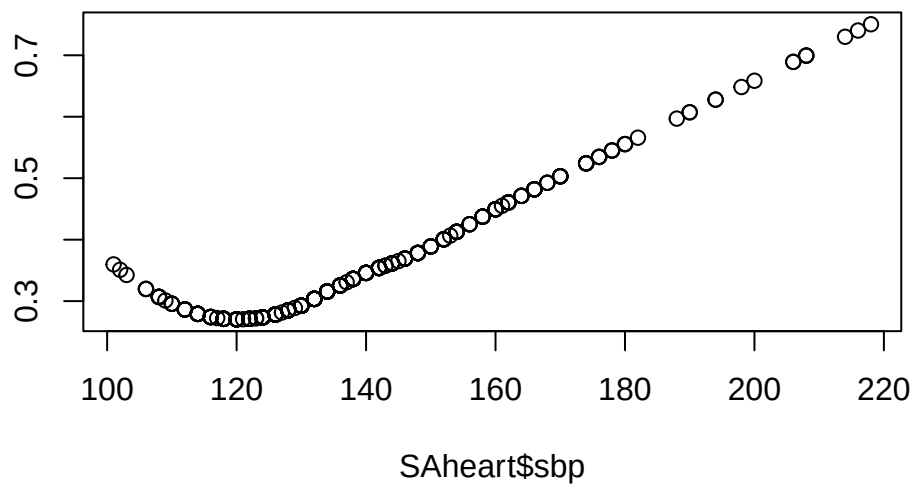
Question 9

We have throughout this practical implicitly taken the loess smoother estimate as if were ‘correct’. It is though just an estimate, and in particular the estimate it gives is affected by the bandwidth, which is specified by the `span` argument to `loess`. Examine the documentation for `loess` and re-run it with different values of `span` to see how the value affects the loess estimate of the probability of CHD by SBP.

SOLUTIONS

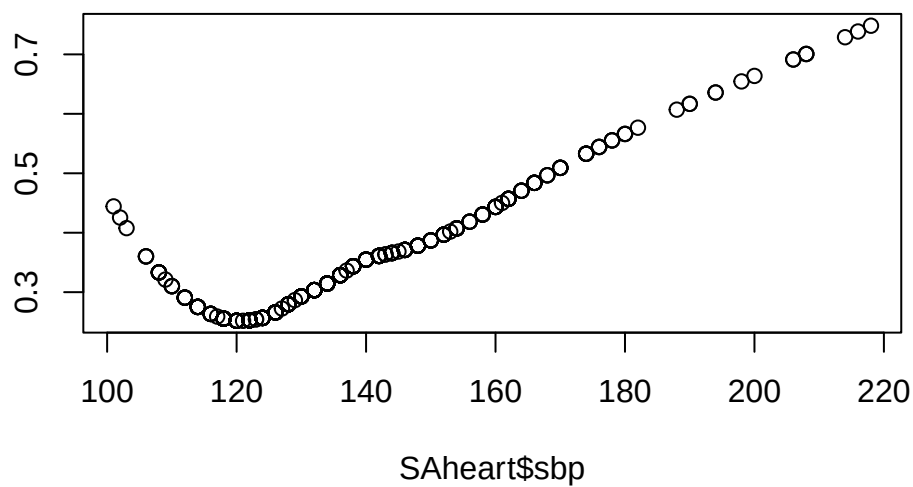
```
plot(SAheart$sbp,  
     predict(loess(chd~sbp, data=SAheart, span=0.9)))
```

redict(loess(chd ~ sbp, data = SAheart, span = (

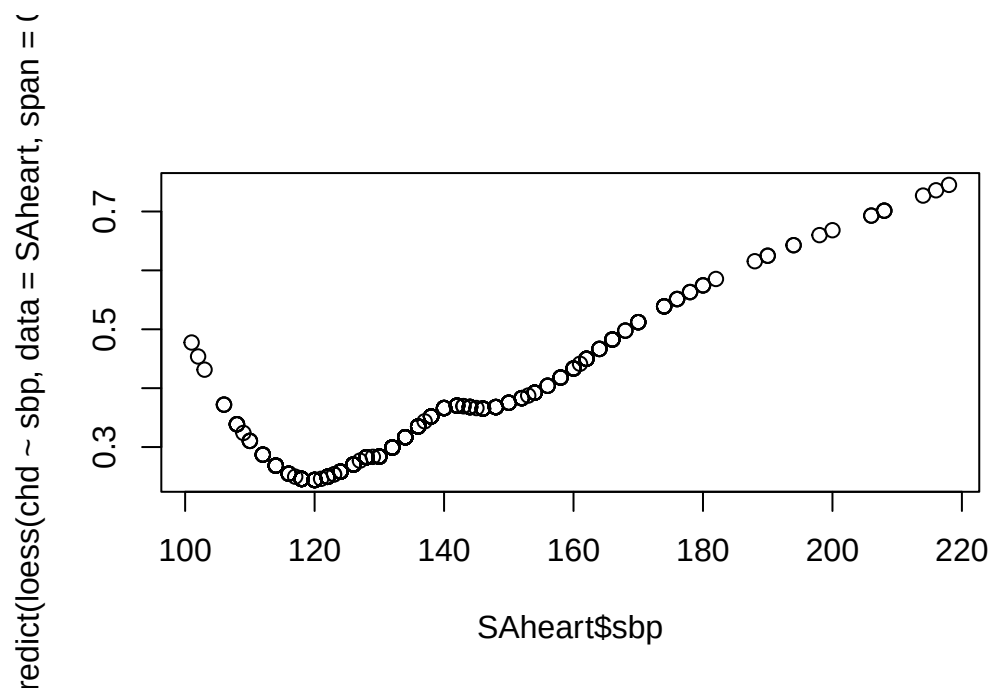


```
plot(SAheart$sbp,  
      predict(loess(chd~sbp, data=SAheart, span=0.75)))
```

edict(loess(chd ~ sbp, data = SAheart, span = 0

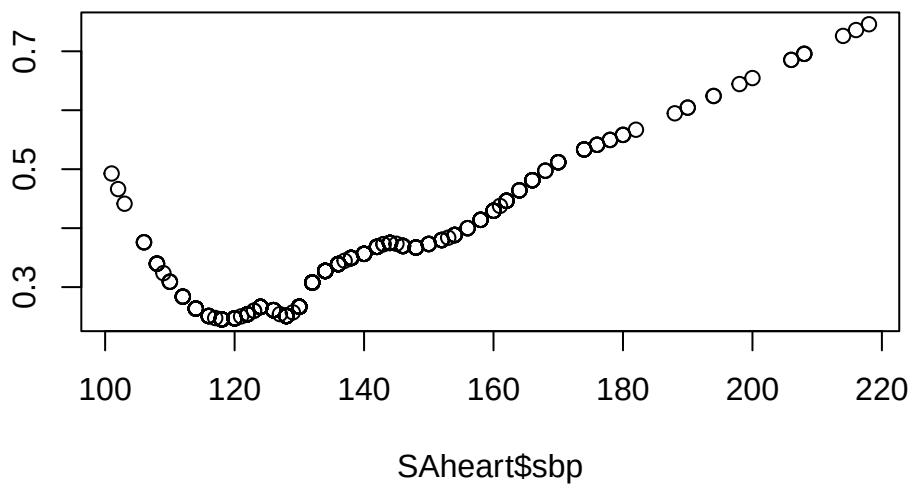


```
plot(SAheart$sbp,  
     predict(loess(chd~sbp, data=SAheart, span=0.6)))
```



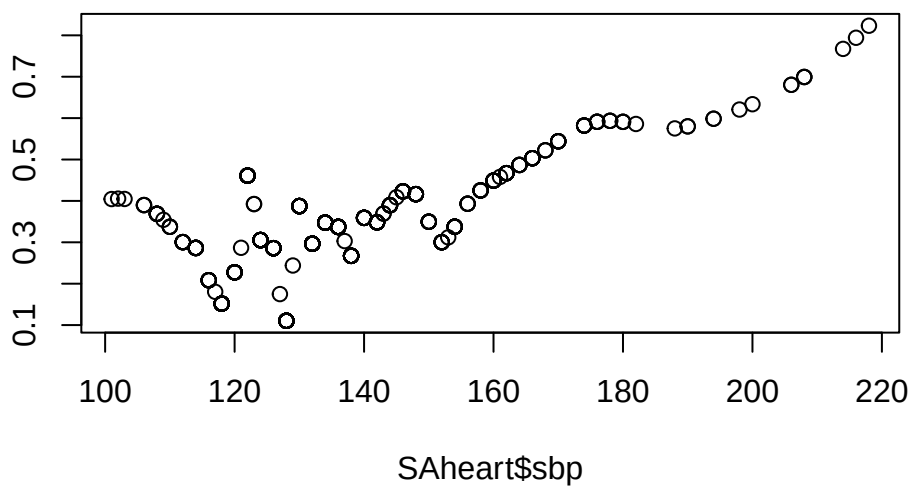
```
plot(SAheart$sbp,  
     predict(loess(chd~sbp, data=SAheart, span=0.5)))
```

redict(loess(chd ~ sbp, data = SAheart, span = (



```
plot(SAheart$sbp,  
      predict(loess(chd~sbp, data=SAheart, span=0.2)))
```

redict(loess(chd ~ sbp, data = SAheart, span = (



We see that with `span=0.9` we get a slightly smoother fit than earlier. `span=0.75` is the default and is what we therefore saw earlier. As the `span` value gets smaller, we get an increasingly wiggly fit. Choosing a larger value gives a fit that is smoother and less variable (in repeated samples), at the expense of some bias compared to the true functional form. Conversely smaller values will give less biased estimates of the function, at the expense of higher variability. The latter manifests itself in fitted lines that are wiggly.